- Variational Integrator

  Specialized for Lagrangian System

  discretize / approximate Lagrangian / least action principle

$$\min_q \int_{t_0}^{t_{fin}} L(q,\dot{q})\,dt$$

$$\downarrow$$

$$\min_q \sum_k \int_{t_k}^{t_{k+1}} L(q,\dot{q})\,dt \qquad t_{k+1} = t_k + h$$

$$\downarrow$$

$$\int_{t_k}^{t_{k+1}} L(q,\dot{q})\,dt \approx \underbrace{h \cdot L\left(\frac{q_{k+1}+q_k}{2}, \frac{q_{k+1}-q_k}{h}\right) := L_d(q_k, q_{k+1})}_{\text{discrete Lagrangian}}$$

$$\min_q \sum_k h\,L\left(\frac{q_{k+1}+q_k}{2}, \frac{q_{k+1}-q_k}{h}\right) = \min_q \sum_k L_d(q_k, q_{k+1})$$

$$\frac{\partial}{\partial q_k}\left[L_d(q_{k-1}, q_k) + L_d(q_k, q_{k+1})\right] = 0$$ discrete Euler-Lagrange Equation

given (q_k, q_{k+1}), can solve for q_{k+1} (or backward) with newtons method

produce reasonable behavior for long-term simulation

for a conservative system, energy = constant defines a manifold

RK method can drift away from the manifold

but variational integrator keep close to the manifold

- Legendre Transform & Hamiltonian

1. need q_1 and $\dot{q}_1$ to initialize variational integrator. how to initialize with q_1 and $\dot{q}_1$?
2. How to get $\dot{q}_k$ in general?

$$\min_q \int L(q,\dot{q})\,dt$$

$$\downarrow$$

$$\min_{q,v} \int L(q,v)\,dt$$

$$\text{s.t. } \dot{q} = v$$] kinematic constraints

eg. If q is quaternion, and v is angular velocity

can put a quaternion kinematics here

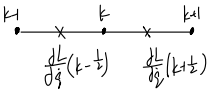
$$\begin{cases} \frac{\partial}{\partial q}\int L(q,v) + p^T(\dot{q}-v)\,dt = 0 \\ \frac{\partial}{\partial v}\int L(q,v) + p^T(\dot{q}-v)\,dt = 0 \\ \dot{q} = v \end{cases}$$

$$\begin{cases} \frac{\partial L}{\partial q} - \dot{p} = 0 \\ \frac{\partial L}{\partial v} + p = 0 \\ \dot{q} = v \end{cases} \Rightarrow \frac{\partial L}{\partial \dot{q}} = p$$

$$L_d(q_k, q_{k+1}) = hL\left(\frac{q_{k+1}+q_k}{2}, \frac{q_{k+1}-q_k}{h}\right)$$

$$D_2 L_d(q_k, q_{k+1}) = h\left[\frac{\partial L}{\partial q}\cdot\frac{1}{2} + \frac{\partial L}{\partial \dot{q}}\cdot\frac{1}{h}\right] = \frac{\partial L}{\partial q}(k+\tfrac{1}{2}) + \frac{h}{2}\frac{\partial L}{\partial q}(k+\tfrac{1}{2})$$

$$-D_1 L_d(q_k, q_{k+1}) = -h\left[\frac{\partial L}{\partial q}\cdot\frac{1}{2} - \frac{\partial L}{\partial \dot{q}}\cdot\frac{1}{h}\right] = \frac{\partial L}{\partial q}(k+\tfrac{1}{2}) - \frac{h}{2}\frac{\partial L}{\partial q}(k+\tfrac{1}{2})$$



$$\frac{d}{dq_k}\left[L_d(q_{k-1}, q_k) + L_d(q_k, q_{k+1})\right] = 0$$

$$\frac{\partial L}{\partial \dot{q}}(k-\tfrac{1}{2}) - \frac{\partial L}{\partial \dot{q}}(k+\tfrac{1}{2}) = \alpha(h)$$

Let $L = \frac{1}{2}v^T M(q) v - U(q)$

$$\begin{cases} \frac{\partial L}{\partial q} - \dot{p} = 0 \\ \frac{\partial L}{\partial v} - p = M(q)v - p = 0 \Rightarrow p = M(q)v \\ \dot{q} = v \end{cases}$$

the dual variable is actually the momentum

⇓ replace v with $M(q)^{-1}p$

(eliminate one primal variable)

* Initialize the variational integrator q_0 and v_0

compute $p_0 = M(q_0)v_0$

find q_1 s.t. $-D_1 L_d(q_0, q_1) = p_0$.

* can also get velocity $v_k = M(q_k)^{-1} p_k$

$$L = \int \frac{1}{2} v^T M(q) v - U(q) + p^T(\dot{q} - v)\, dt$$

$$= \int \frac{1}{2} p^T M(q)^{-1} p - U(q) + p^T(\dot{q} - M(q)^{-1}p)\, dt$$

$$= \int -\frac{1}{2} p^T M(q)^{-1} p - U(q) + p^T \dot{q}\, dt$$

$H(p,q) = \frac{1}{2}p^T M(q)^{-1} p + U(q)$ = kinetic energy + potential energy

the dual is $\max_p \min_q \int -H(p,q) + p^T \dot{q}\, dt$

$$\begin{cases} \frac{d}{dq}\int H(p,q) - p^T\dot{q}\, dt = 0 \\ \frac{d}{dp}\int H(p,q) - p^T\dot{q}\, dt = 0 \end{cases} \Rightarrow \boxed{\begin{cases} \frac{\partial H}{\partial q} + \dot{p} = 0 \\ \frac{\partial H}{\partial p} - \dot{q} = 0 \end{cases}}$$ Hamiltonian's Equation

Given q_0, v_0

compute $p_0 = M(q_0)v_0$

loop $k = 0, 1, \cdots$

find q_{k+1} s.t. $-D_1 L_d(q_k, q_{k+1}) = p_k$

$p_{k+1} = D_2 L_d(q_k, q_{k+1})$

$v_{k+1} = M(q_k)^{-1} p_{k+1}$ if needed

find q_{k+1} s.t. $D_2 L_d(q_{k-1}, q_k) = -D_1 L_d(q_k, q_{k+1})$

DEL with Constraint

find q s.t. $C(q) = 0$ $\quad \frac{d}{d\varepsilon}\int L(q, \dot{q})\, dt = \int \left(\frac{\partial L}{\partial q} - \frac{d}{dt}\frac{\partial L}{\partial \dot{q}}\right) \delta q\, dt = 0$ $\forall$ admissible δq

find q s.t. $C(q_{k+1}) = 0$ $\quad \frac{d}{d\varepsilon}\sum L_d(q_k, q_{k+1})\, \delta q = 0$ $\forall$ $\frac{\partial C}{\partial q_k}\delta q_k = 0$

$$\begin{cases} \frac{\partial}{\partial q_k} L_d(q_{k-1}, q_k) + \frac{\partial}{\partial q_k} L_d(q_k, q_{k+1}) = \frac{\partial C}{\partial q_k}^T \lambda \\ C(q_{k+1}) = 0 \end{cases}$$

- DEL with non-conservative Force

$$\frac{d}{dq}\int L(q,\dot{q})\,dt = \int\left(\frac{\partial L}{\partial q} - \frac{d}{dt}\frac{\partial L}{\partial \dot{q}}\right)\delta q\,dt = -\int_{t_0}^{t_N}\sum_{i=1}^{N} F_{i,nc}^{(act)^T} Dr_i(q)\,\delta q\,dt$$

$$\frac{d}{dq}\sum_{k}^{} L_d(q_k, q_{k+1})\,\delta q + \sum_{k}^{}\sum_{i=1}^{N} h\,F_{i,nc}^{(act)}(k+\tfrac{1}{2})\,Dr_i\left(\frac{q_k+q_{k+1}}{2}\right)\delta q = 0$$

$$\frac{d}{dq_k}L_d(q_{k-1}, q_k) + \frac{d}{dq_k}L_d(q_k, q_{k+1}) + h\sum_{i=1}^{N}\left[Dr_i\left(\frac{q_{k-1}+q_k}{2}\right)^T F_{i,nc}^{(act)}(k-\tfrac{1}{2}) + Dr_i\left(\frac{q_k+q_{k+1}}{2}\right) F_{i,nc}^{(act)}(k+\tfrac{1}{2})\right] = 0$$

note: force can be a function of q and $\dot{q}$ (eg. PD controller)

- HJB Equation

$$\min_{u[0,T]} \int_0^T \ell(x(s), u(s))\,ds + \phi(x(T))$$

$$\text{s.t.} \quad \dot{x} = f(x,u) \qquad x(0) = x_0$$

the value function (optimal cost-to-go)

$$V(x,t) = \inf_{u[t,T]} \int_t^T \ell(x(s), u(s))\,ds + \phi(x(T))$$

$$\text{s.t.} \quad \dot{x} = f(x,u) \qquad x(t) = x$$

Dynamic programming Principle

$$V(x,t) = \inf_{u[t,t+\delta t]} \int_t^{t+\delta t} \ell(x(s), u(s))\,ds + V(x(t+\delta t), t+\delta t)$$

$$= \inf_{u(t)} \; \text{s.t.}\; \ell(x(t), u(t))\,\delta t + O(\delta t) + V(x(t), t) + \delta t\, f(x(t), u(t))^T \nabla_x V(x(t), t) + \delta t \frac{\partial}{\partial t} V(x(t), t) + O(\delta t)$$

$$\text{s.t.}\; \ell(x(t), u(t))\,\delta t + \delta t\, f(x(t), u(t))^T \nabla_x V(x(t), t) + \delta t \frac{\partial}{\partial t} V(x(t), t) + O(\delta t)$$

HJB Equation

$$\min_{u(\cdot):[0,T]} \int_0^T \ell(x(s),u(s))\,ds + \phi(x(T))$$

$$s.t \quad \dot{x} = f(x,u) \qquad x(0) = x_0$$

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Dynamic programming principle

$$V(x,t) = \inf_{u(\cdot):[t,t+\delta t]} \int_t^{t+\delta t} \ell(x(s),u(s))\,ds + V(x(t+\delta t), t+\delta t)$$

$$= \inf_{u\in\mathcal{U}} \left[\delta t\, \ell(x(t),u(t)) + o(\delta t) + V(x(t),t) + \nabla_x V(x(t),t)^T f(x(t),u(t))\cdot \delta t + \frac{\partial}{\partial t} V(x(t),t)\delta t + o(\delta t) \right]$$

$$\inf_{u\in\mathcal{U}} \left\{ \delta t\, \ell(x(t),u(t)) + \delta t\, \nabla_x V(x(t),t)^T f(x(t),u(t)) + \frac{\partial}{\partial t} V(x(t),t)\,\delta t + o(\delta t) \right\} = 0$$

$$\begin{cases} -\frac{\partial}{\partial t} V(x(t),t) = \inf_u \left\{ \ell(x(t),u) + \nabla_x V(x(t),t)^T f(x(t),u(t)) \right\} & t < T \\ V(x,T) = \phi(x) \end{cases}$$

Define $H(p;x) = \inf_u \left\{ \ell(x,u) + p^T f(x,u) \right\}$ Concave

$$\begin{cases} \frac{\partial}{\partial t} V(x(t),t) + H\left(\nabla_x V(x(t),t), x(t)\right) = 0 \\ V(x,T) = \phi(x) \end{cases}$$